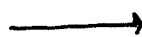


3.3

$$1. \{z \mid \operatorname{Re} z < 0, |\operatorname{Im} z| < \pi\} \xrightarrow{e^z} \mathbb{D} \setminus (-1, 0]$$

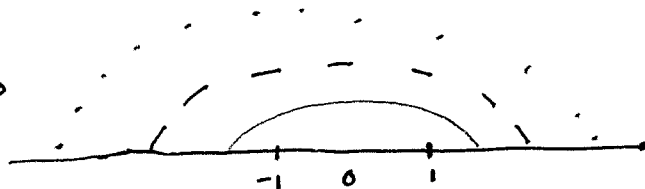
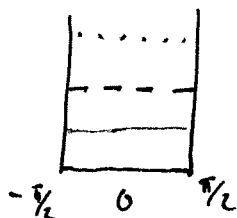


$$3a) \sin(-z) = -\sin z, \quad \sin \bar{z} = \overline{\sin z}, \quad \sin(\pi+z) = -\sin z, \quad \sin(z+2\pi) = \sin z$$

$$(\sin z)' = \cos z \Rightarrow (\sin z)' = 0 \text{ for } z = (2k+1)\frac{\pi}{2}, k \in \mathbb{Z}$$

$\Rightarrow$ need only consider $HVS_s = \{z \mid \operatorname{Im} z > 0, |\operatorname{Re} z| < \frac{\pi}{2}\}$

$$HVS_s \xrightarrow{\sin z} UHP$$

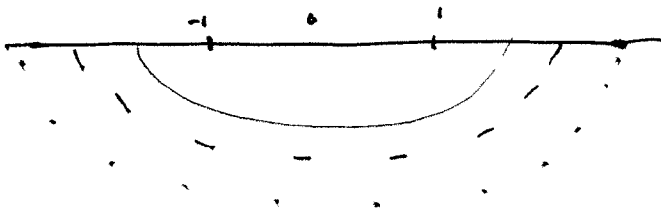
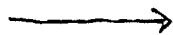
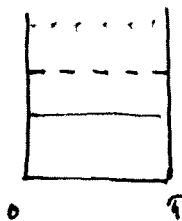


$$b) \cos(-z) = \cos z, \quad \cos \bar{z} = \overline{\cos z}, \quad \cos(\pi+z) = -\cos z, \quad \cos(z+2\pi) = \cos z$$

$$(\cos z)' = -\sin z \Rightarrow (\cos z)' = 0 \text{ for } z = k\pi, k \in \mathbb{Z}$$

$\Rightarrow$ need only consider $HVS_c = \{z \mid \operatorname{Im} z > 0, 0 < \operatorname{Re} z < \pi\}$

$$HVS_c \xrightarrow{\cos z} LHP$$



5. Fixed Points

Dilation $Tz = az, a \neq 1$: $0, \infty$

Translation $Tz = z + a, a \neq 0$: ∞

Inversion $Tz = \frac{1}{z}$: $1, -1$

8 $Tz = \frac{az+b}{cz+d}$ Möbius $T(\mathbb{R}_\infty) = \mathbb{R}_\infty \iff \exists \text{ choice } a, b, c, d \in \mathbb{R}$

$\Leftarrow T(0) = \frac{b}{d}, T(1) = \frac{a+b}{c+d}, T(-1) = \frac{-a+b}{-c+d}$ each belong to $\mathbb{R}_\infty$

hence $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$

$\Rightarrow \exists x_2 \in \mathbb{R}_\infty \rightarrow T(x_2) = 1, x_3 \in \mathbb{R}_\infty \rightarrow T(x_3) = 0, x_4 \in \mathbb{R}_\infty \rightarrow T(x_4) = \infty$

hence $Tz = (z, x_2, x_3, x_4) = \frac{x_2 - x_4}{x_2 - x_3} \frac{z - x_3}{z - x_4}$

$$= \frac{(x_2 - x_4)z + (x_2 - x_4)(-x_3)}{(x_2 - x_3)z + (x_2 - x_3)(-x_4)} = \frac{az + b}{cz + d}$$

and by construction $a, b, c, d \in \mathbb{R}_\infty$

Note if any one x_2, x_3, x_4 is ∞ appropriately modify Tz

10 $T: \mathbb{D} \rightarrow \mathbb{D} \Rightarrow \exists a \in \mathbb{D} \rightarrow Ta = 0$

Case 1 $a = 0 \Rightarrow T0 = 0$ & $T\infty = \infty \Rightarrow Tz = \alpha z$ & $|\alpha| = 1$

Case 2 $a \neq 0 \Rightarrow Ta = 0$ & $Ta^* = \infty$ ($a^* = \frac{1}{\bar{a}}$)

$$\Rightarrow Tz = \lambda \frac{z-a}{z-a^*} = -\lambda \bar{a} \frac{z-a}{1-\bar{a}z}$$

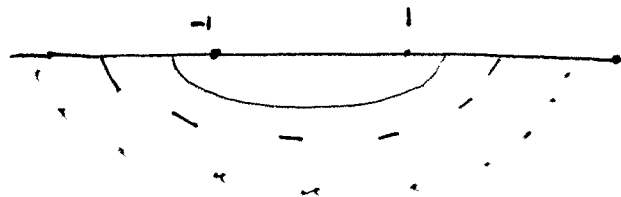
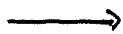
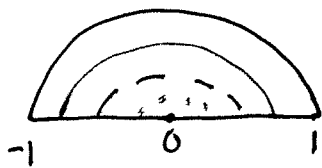
$$|T1| = 1 \Rightarrow |-\lambda \bar{a}| = 1$$

$$Tz = \alpha \frac{z-a}{1-\bar{a}z} \text{ & } |\alpha| = 1$$

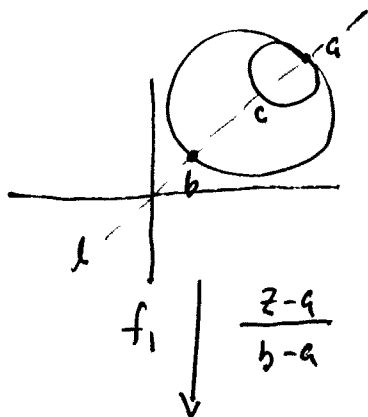
13. $f(z) = f\left(\frac{1}{z}\right)$, $f(\bar{z}) = \overline{f(z)}$, $f(-z) = -f(z)$

$\Rightarrow$ need only consider $D_+ = \{z \in D \mid \text{Im } z > 0\}$

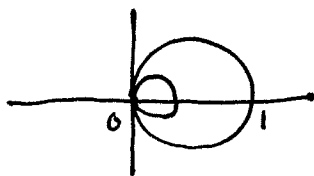
$D_+ \xrightarrow{f} \text{LHP}$



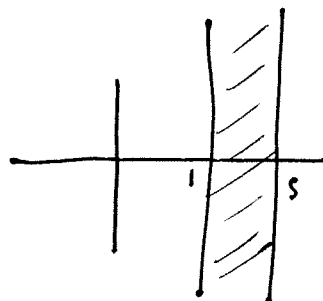
14.



let l be line of symmetry and let b be the other intersection point of l with outer circle

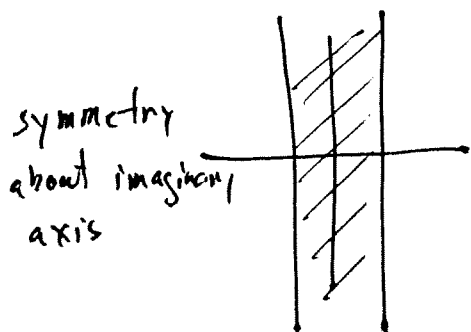


$f_2 \xrightarrow{\frac{1}{z}}$

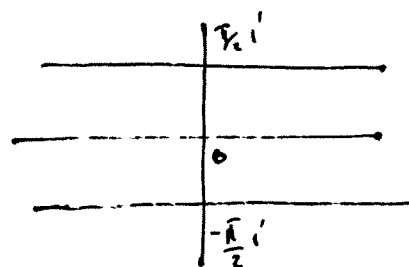


$s = f_2(f_1(c))$

$f_3 \swarrow \left(z - \frac{s+1}{2}\right)$

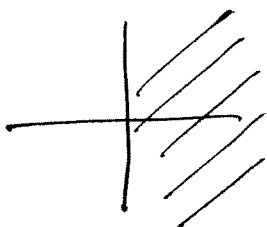


$f_4 \xrightarrow{i \times z}$

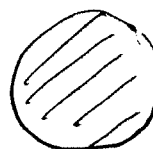


$\alpha = \frac{\frac{\pi}{2}}{\frac{s-1}{2}}$

$f_5 \swarrow e^z$

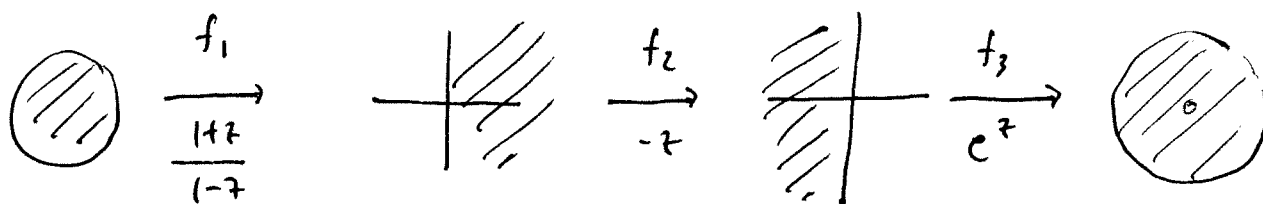


$f_6 \xrightarrow{\frac{z-1}{z+1}}$



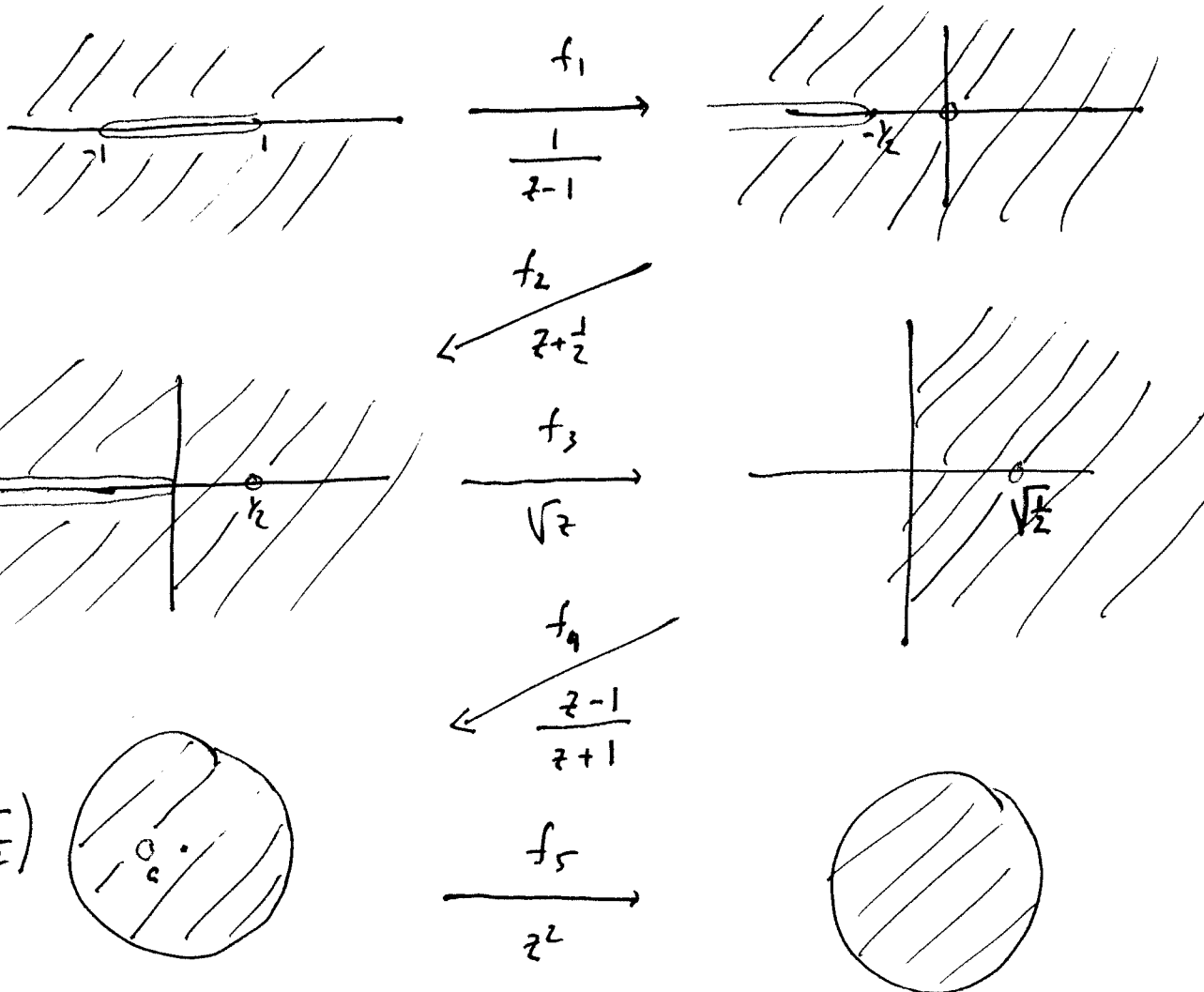
$f = f_6 \circ f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1$

15



$$f = f_3 \circ f_2 \circ f_1$$

16



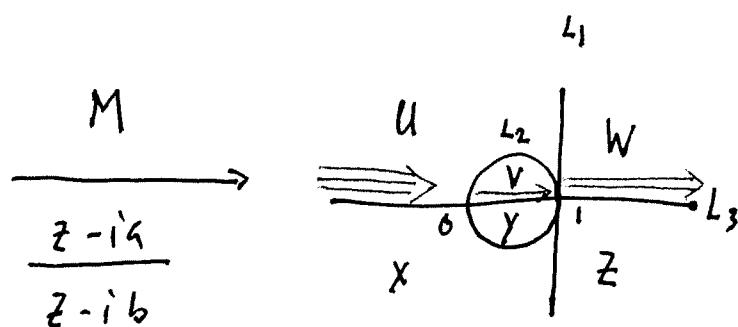
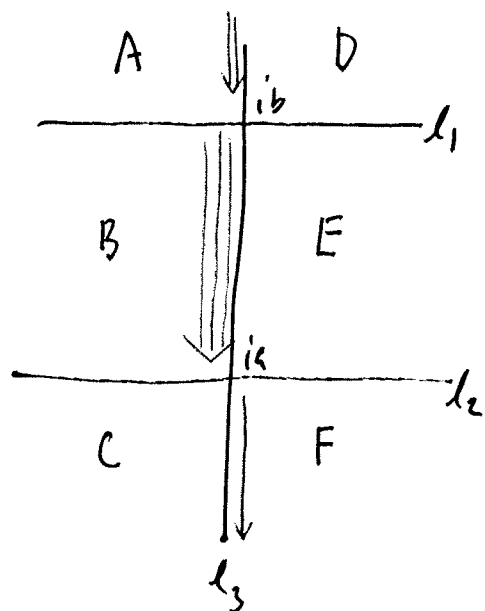
$$a = f_4\left(\sqrt{\frac{1}{2}}\right)$$

$$a \neq 0$$

$$f = f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1$$

17. Suppose $f \in \text{Aut}(G)$, G region $\rightarrow f(G) \subset \mathbb{C}$ a circle
 Let c be center of $\mathbb{C}$ and $f_2(z) = z - c$. Let $g = f_2 \circ f$
 Then $g(G)$ has constant modulus $\Rightarrow g$ constant (in class)
 $\Rightarrow f$ constant.

18



$$M(l_j) = L_j$$

orient L_3 as

$$ia \longrightarrow \infty$$

$$\infty \implies ib$$

$$ib \implies ia$$

then L_3 is oriented as

$$0 \longrightarrow 1$$

$$1 \implies \infty$$

$$\infty \implies 0$$

$$\therefore M(F) = V$$

$$M(C) = Y$$

$$M(D) = W$$

$$M(A) = Z$$

$$M(E) = U$$

$$M(B) = X$$

20

$$S_z = T_z$$

ff

$$\exists \lambda \rightarrow$$

$$\begin{aligned} \alpha &= ax \\ \beta &= bx \\ \gamma &= cy \\ \delta &= dx \end{aligned}$$

 $\Leftarrow$ trivial

$$\Rightarrow S_z = \left(z, \frac{b-d}{a-c}, \frac{-b}{a}, \frac{-d}{c} \right)$$

$$T_z = \left(z, \frac{\beta-\delta}{\alpha-\gamma}, \frac{-\beta}{\alpha}, \frac{-\delta}{\gamma} \right)$$

$$(1) \quad \frac{-b}{a} = \frac{-\beta}{\alpha} \Rightarrow \frac{\alpha}{a} = \frac{\beta}{b} \Rightarrow \exists x \rightarrow \begin{aligned} \alpha &= ax \\ \beta &= bx \end{aligned}$$

$$(2) \quad \frac{-d}{c} = \frac{-\delta}{\gamma} \Rightarrow \frac{\gamma}{c} = \frac{\delta}{d} \Rightarrow \exists y \rightarrow \begin{aligned} \gamma &= cy \\ \delta &= dy \end{aligned}$$

$$(3) \quad \frac{b-d}{a-c} = \frac{\beta-\delta}{\alpha-\gamma} = \frac{bx-dy}{ax-cy}$$

$$(b-d)(ax-cy) = (bx-dy)(a-c) \Rightarrow x=y$$

$$22. \quad T(z) = z \Rightarrow \begin{aligned} &\text{exactly} \\ &z = 0, \infty \end{aligned} \quad \text{ff} \quad T(z) = \alpha z \quad \alpha \neq 1$$

 $\Leftarrow$ trivial

$$\Rightarrow \begin{aligned} T_0 = 0 &\Rightarrow b=0 \\ T_\infty = \alpha &\Rightarrow c=0 \end{aligned} \Rightarrow Tz = \frac{a}{d} z = \alpha z \quad \alpha \neq 1$$

$$23. \quad T_0 = \infty, \quad T_{\infty} = 0 \quad \text{ff} \quad Tz = \frac{\alpha}{z}, \quad \alpha \neq 0$$

 $\Leftarrow$ Trivial

$$\Rightarrow \begin{aligned} T_\infty = 0 &\Rightarrow a=0 \\ T_0 = \infty &\Rightarrow d=0 \end{aligned}$$

$$T(z) = \frac{b}{c} \frac{1}{z} = \frac{\alpha}{z}, \quad \alpha \neq 0$$

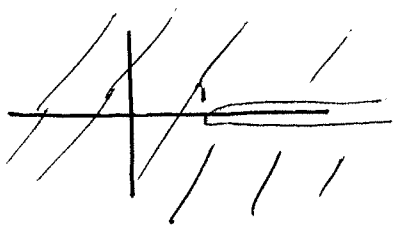
28

$$Tz = (1-z)^i = \exp^{i \log(1-z)}$$

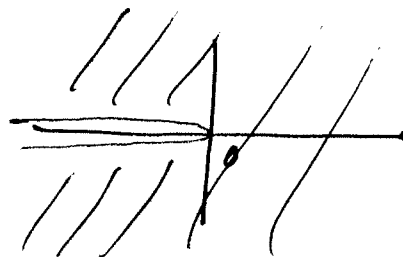
where $\log$ is principal branch of $\log z$

domain of T is $\mathbb{C} \setminus [1, \infty)$

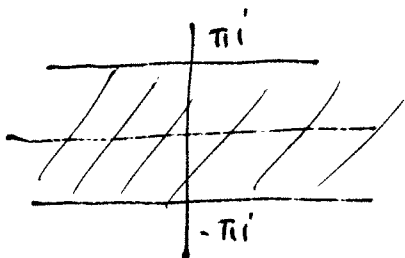
$$Tz = e^{-\arg(1-z) + i \log|1-z|}$$



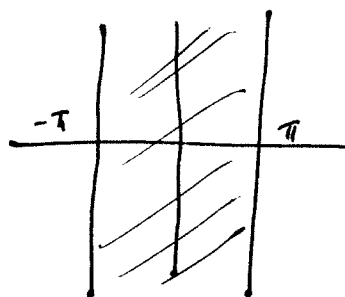
$$\xrightarrow[f_1]{1-z}$$



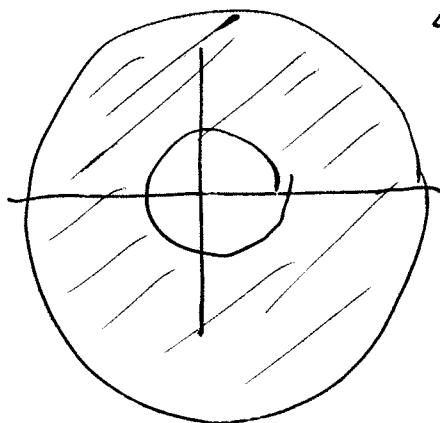
$$\xleftarrow[f_2]{\log z}$$



$$\xrightarrow[f_3]{iz}$$

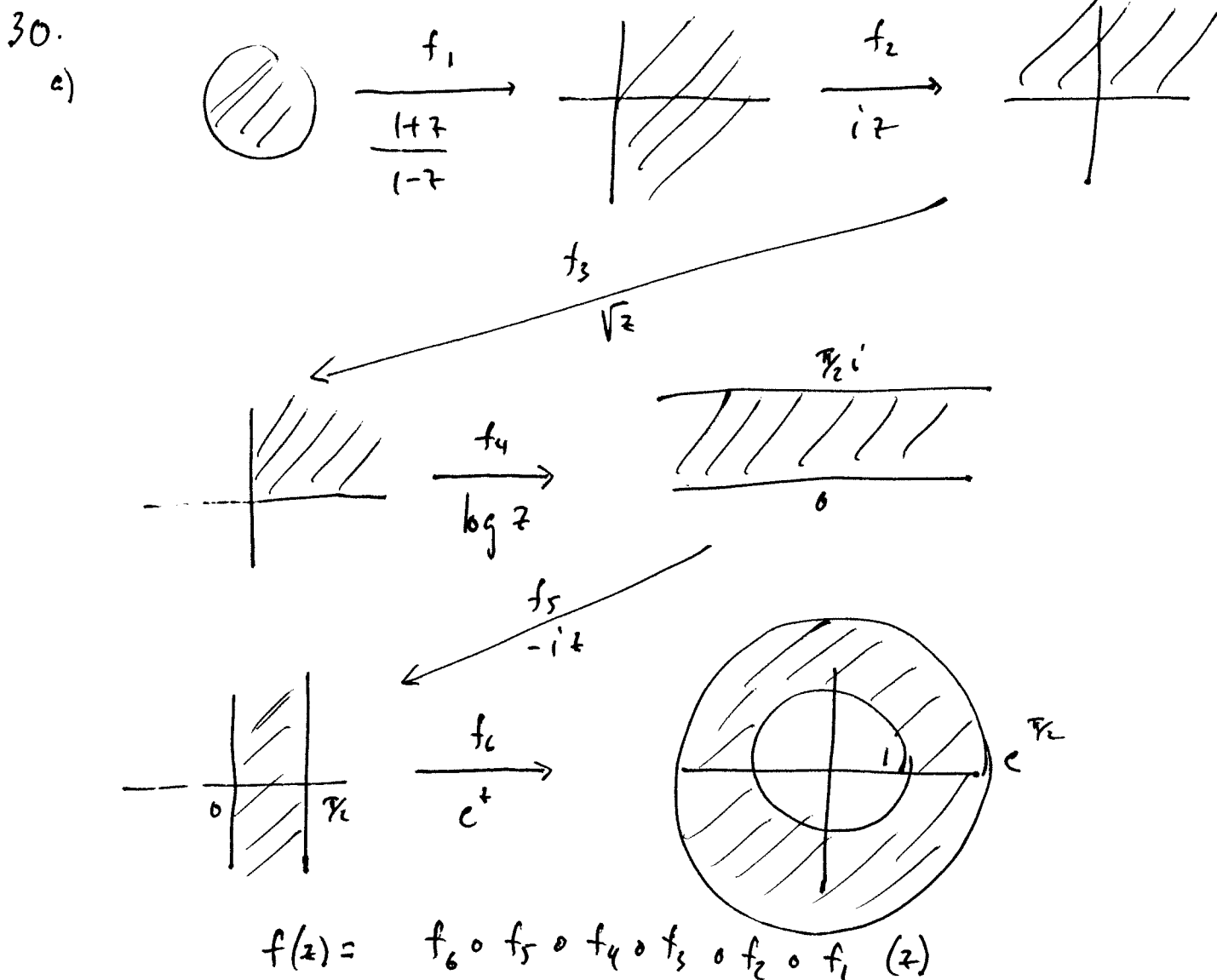


$$\xleftarrow[f_4]{e^z}$$



inner radius $e^{-\pi}$
outer radius e^{π}

$$T = f_4 \circ f_3 \circ f_2 \circ f_1$$



b) Suppose $f(S(z)) = f(z) \Rightarrow$

$$\exp \left\{ -i \log \left[i \frac{1+s}{1-s} \right]^{\gamma_2} \right\} = \exp \left\{ -i \log \left[i \frac{1+z}{1-z} \right]^{\gamma_2} \right\} \Rightarrow$$

$$-i \log \left[i \frac{1+s}{1-s} \right]^{\gamma_2} = -i \log \left[i \frac{1+z}{1-z} \right]^{\gamma_2} + 2\pi ki \quad \text{for some } k \in \mathbb{Z} \Rightarrow$$

$$\left[i \frac{1+s}{1-s} \right]^{\gamma_2} = \left[i \frac{1+z}{1-z} \right]^{\gamma_2} e^{-2\pi k} \Rightarrow \frac{1+s}{1-s} = \frac{1+z}{1-z} e^{-4\pi k} \Rightarrow$$

$$s = \frac{w-1}{w+1} \quad \text{where } w = \frac{1+z}{1-z} e^{-4\pi k}$$

P.1. Given $\varepsilon > 0 \quad \exists N \rightarrow n > N \Rightarrow e^{-\frac{1}{n}} < \frac{\varepsilon}{4}$

Let $\delta_1 = \frac{1}{N+1}$. Then for $z, w \in \overline{B(0, \delta_1)} \setminus \{0\}$

we have $|f(z) - f(w)| < \frac{\varepsilon}{2}$.

$\overline{B(0, 1)} \setminus B(0, \delta_1)$ is compact and f is continuous on it.

By Thm 5.15 f is unif. cont on $\overline{B(0, 1)} \setminus B(0, \delta_1)$; i.e.,

given $\varepsilon > 0 \quad \exists \delta_2 \rightarrow |z - w| < \delta_2 \Rightarrow |f(z) - f(w)| < \frac{\varepsilon}{2}$

$\therefore$ On $B(0, 1) \setminus \{0\}$ given $\varepsilon > 0$ choose $\delta = \min(\delta_1, \delta_2)$.

P.2. $e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$ converges absolutely $\forall z$

Case 1 $z > 0$, then $\frac{z^2}{2!} + \frac{z^3}{3!} + \dots > 0 \Rightarrow e^z > 1 + z$

Case 2 $z \leq -1$, then $1 + z \leq 0 < e^z$

Case 3 $-1 < z < 0$ Let $x = -z$ so that $0 < x < 1$

$$e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{3!} + \dots$$

but $\frac{x^2}{2} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$ is (1) alternating
(2) decreasing i.e.,
 $\frac{x^n}{n!} > \frac{x^{n+1}}{(n+1)!} \quad \forall n$

$$\therefore \frac{x^2}{2} - \frac{x^3}{3!} + \frac{x^4}{4!} + \dots > 0$$

$$e^{-x} > 1 - x \quad (\Leftrightarrow) \quad e^z > 1 + z$$

P. 3. a) $\cos z = 3i$ $z = x+iy$

$$\cos 2x \cosh 2y - i \sin 2x \sinh 2y = 3i \Rightarrow$$

$$\cos 2x \cosh 2y = 0 \quad \& \quad -\sin 2x \sinh 2y = 3$$

$$\cosh s \neq 0 \text{ for reals}$$

$$\therefore \cos 2x = 0$$

$$\Rightarrow 2x = \text{odd mult } \frac{\pi}{2} = (2k+1)\frac{\pi}{2} \Rightarrow \sin 2x = \pm 1$$

$$2x = (4k+1)\frac{\pi}{2} \Rightarrow \sin 2x = +1 \Rightarrow \sinh 2y = -3$$

$$\Rightarrow y = \frac{1}{2} \sinh^{-1}(-3) \\ = -\frac{1}{2} \sinh^{-1}(3)$$

$$2x = (4k+3)\frac{\pi}{2} \Rightarrow \sin 2x = -1$$

$$\Rightarrow \sinh 2y = 3$$

$$\Rightarrow y = \frac{1}{2} \sinh^{-1} 3$$

b) $\sin z = 8$ $z = x+iy$

$$\sin x+iy = \sin x \cosh y + i \cos x \sinh y = 8 \Rightarrow$$

$$\sin x \cosh y = 8 \quad \& \quad \cos x \sinh y = 0$$

$$\& \sinh y = 0 \Leftrightarrow y = 0, \text{ then } \cosh y = 1 \text{ and } \sin x = 8 \times$$

$$\therefore \cos x = 0 \Rightarrow x \text{ odd mult } \frac{\pi}{2} \quad x = (2k+1)\frac{\pi}{2}$$

$$\Rightarrow \sin x = \pm 1$$

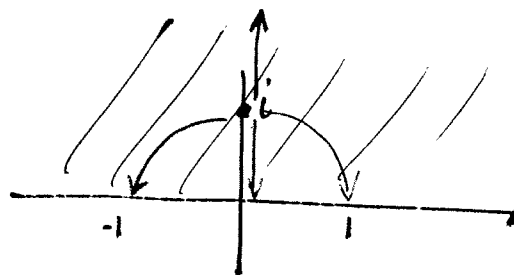
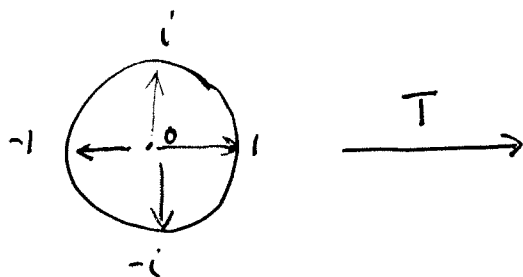
$$x = (4k+1)\frac{\pi}{2} \Rightarrow \sin x = 1 \Rightarrow \cosh y = 8$$

$$\Rightarrow y = \pm \cosh^{-1} 8$$

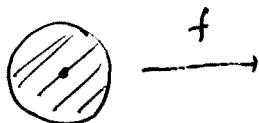
$$x = (4k+3)\frac{\pi}{2} \Rightarrow \sin x = -1 \Rightarrow \cosh y = -8 \times$$

P. 4

$$Tz = \frac{i+1}{i-1} \frac{z-1}{z+1} = -i \frac{z-1}{z+1} = i \frac{1-z}{1+z}$$



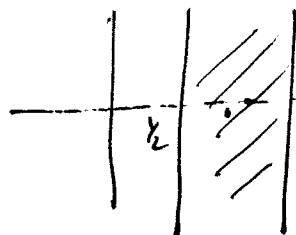
P. 5.



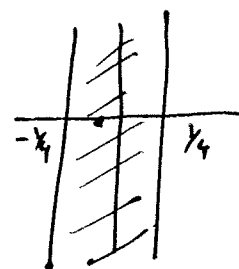
$$f = F^{-1}$$



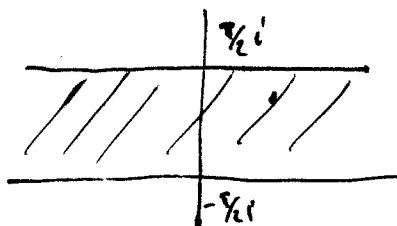
$$\xrightarrow[\frac{1}{z+1}]{F_1}$$



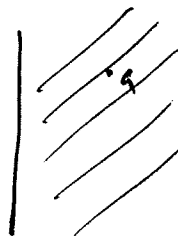
$$\xrightarrow[\frac{z-3/4}{z-1/4}]{F_2}$$



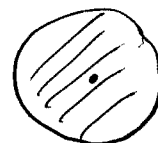
$$\xleftarrow[-2\pi i z]{F_3}$$



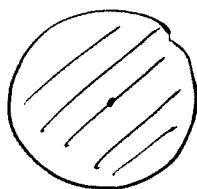
$$\xrightarrow[e^z]{F_4}$$



$$\xrightarrow[\frac{z-a}{z-a^2}]{F_5}$$



$$\xleftarrow[\gamma z]{F_6} \quad |y|=1$$



$$F_1\left(\frac{1}{2}\right) = \frac{2}{3} \quad F_2\left(\frac{2}{3}\right) = -\frac{1}{12} \quad F_3\left(-\frac{1}{12}\right) = \frac{\pi}{6}i$$

$$F_4\left(\frac{\pi}{6}i\right) = e^{\frac{\pi}{6}i} = a \quad F_5(a) = 0 \quad F_6(0) = 0$$

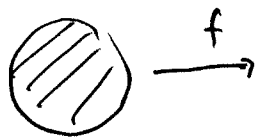
$$F = F_6 \circ F_5 \circ F_4 \circ F_3 \circ F_2 \circ F_1$$

choose $\gamma = e^{i\theta}$ so that

$$F'\left(\frac{1}{2}\right) = F_6'(0) F_5'(e^{\frac{\pi}{6}i}) F_4'\left(\frac{\pi}{6}i\right) F_3'\left(-\frac{1}{12}\right) F_2'\left(\frac{2}{3}\right) F_1'\left(\frac{1}{2}\right)$$

has argument 0

P. 6.



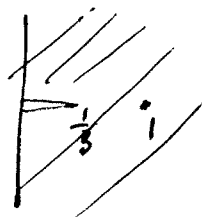
$\xrightarrow{f}$



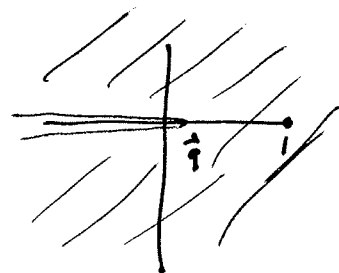
$$f = f^{-1}$$



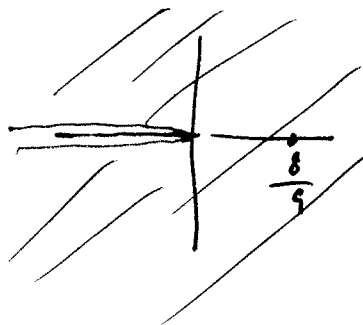
$$\xrightarrow{\frac{F_1}{\frac{1+z}{1-z}}}$$



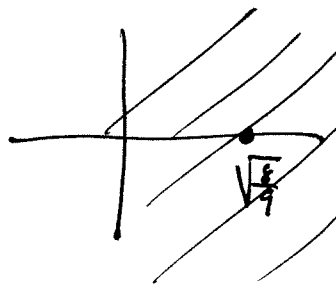
$$\xrightarrow{\frac{F_2}{z^2}}$$



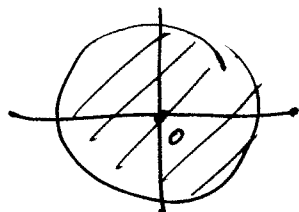
$$\xleftarrow{\frac{F_3}{z - \frac{1}{9}}}$$



$$\xrightarrow{\frac{F_4}{\sqrt{z}}}$$



$$\xleftarrow{\frac{F_5}{\frac{z - \sqrt{\frac{k}{9}}}{z + \sqrt{\frac{k}{9}}}}}$$



$$F = F_5 \circ F_4 \circ F_3 \circ F_2 \circ F_1$$